Full Stack Application Deployment on Kubernetes

Grégoire Tournois

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1 Project Goal

The assignment required:

1. deploying the provided fullstackapp to a Kubernetes cluster;
2. making the application more resilient by adding replicas;
3. changing the frontend message, rebuilding the image, and updating the pod;

2 Building Docker Images

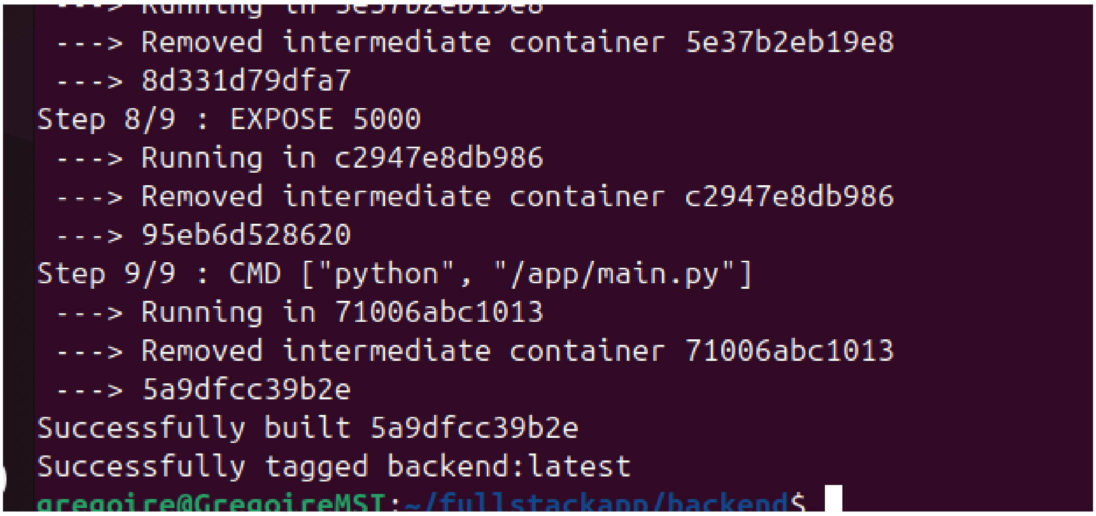
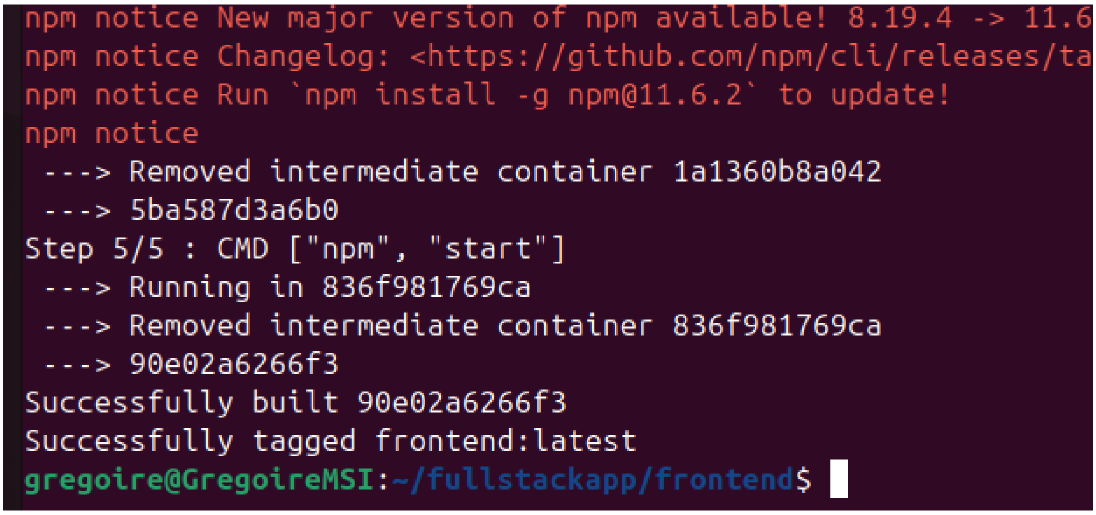


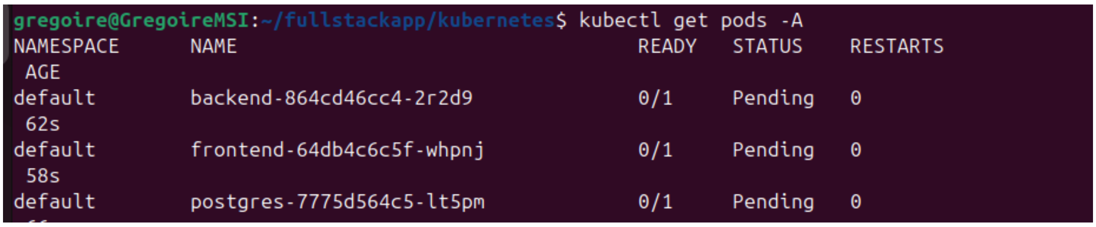
Figure 1: Building the backend image



```
npm notice New major version of npm available! 8.19.4 -> 11.6
npm notice Changelog: <https://github.com/npm/cli/releases/ta
npm notice Run `npm install -g npm@11.6.2` to update!
npm notice
 ---> Removed intermediate container 1a1360b8a042
 ---> 5ba587d3a6b0
Step 5/5 : CMD ["npm", "start"]
 ---> Running in 836f981769ca
 ---> Removed intermediate container 836f981769ca
 ---> 90e02a6266f3
Successfully built 90e02a6266f3
Successfully tagged frontend:latest
gregoire@GregoireMSI:~/fullstackapp/frontend$
```

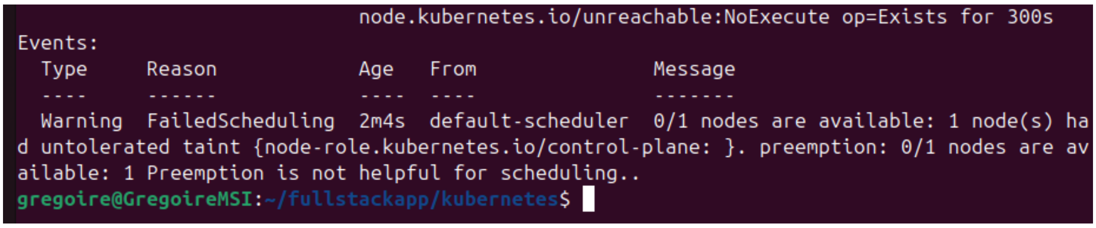
Figure 2: Building the frontend image

3 Initial Scheduling Issue



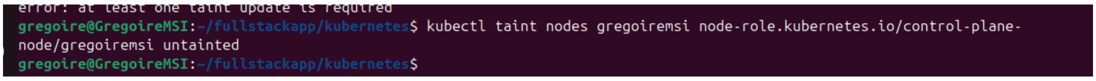
```
gregoire@GregoireMSI:~/fullstackapp/kubernetes$ kubectl get pods -A
NAMESPACE     NAME                             READY   STATUS    RESTARTS
 AGE
default       backend-864cd46cc4-2r2d9         0/1     Pending   0
 62s
default       frontend-64db4c6c5f-whpnj        0/1     Pending   0
 58s
default       postgres-7775d564c5-lt5pm        0/1     Pending   0
```

Figure 3: Pods stuck in Pending after the first deployment



```
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type     Reason            Age   From               Message
  ----     ------            ----  ----               -------
  Warning  FailedScheduling  2m4s  default-scheduler  0/1 nodes are available: 1 node(s) ha
d untolerated taint {node-role.kubernetes.io/control-plane: }. preemption: 0/1 nodes are av
ailable: 1 Preemption is not helpful for scheduling..
gregoire@GregoireMSI:~/fullstackapp/kubernetes$
```

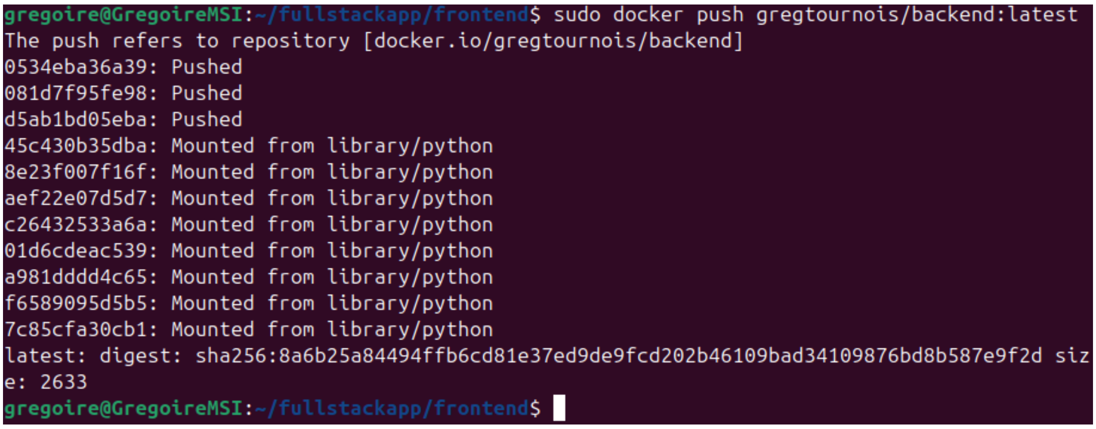
Figure 4: Node shown as tainted, preventing scheduling



```
error: at least one taint update is required
gregoire@GregoireMSI:~/fullstackapp/kubernetes$ kubectl taint nodes gregoiremsi node-role.kubernetes.io/control-plane-
node/gregoiremsi untainted
gregoire@GregoireMSI:~/fullstackapp/kubernetes$
```

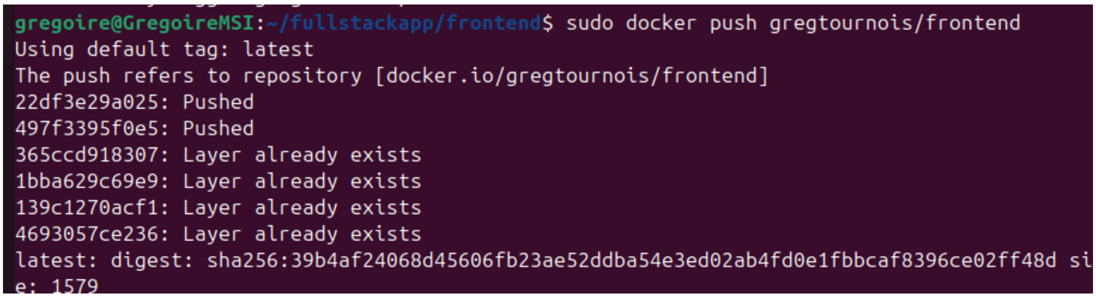
Figure 5: Removing the taint so the pods can be scheduled

4 Pushing Images to solve ErrImageNeverPull



```
gregoire@GregoireMSI:~/fullstackapp/frontend$ sudo docker push gregtournois/backend:latest
The push refers to repository [docker.io/gregtournois/backend]
0534eba36a39: Pushed
081d7f95fe98: Pushed
d5ab1bd05eba: Pushed
45c430b35dba: Mounted from library/python
8e23f007f16f: Mounted from library/python
aef22e07d5d7: Mounted from library/python
c26432533a6a: Mounted from library/python
01d6cdeac539: Mounted from library/python
a981dddd4c65: Mounted from library/python
f6589095d5b5: Mounted from library/python
7c85cfa30cb1: Mounted from library/python
latest: digest: sha256:8a6b25a84494ffb6cd81e37ed9de9fcd202b46109bad34109876bd8b587e9f2d siz
e: 2633
gregoire@GregoireMSI:~/fullstackapp/frontend$
```

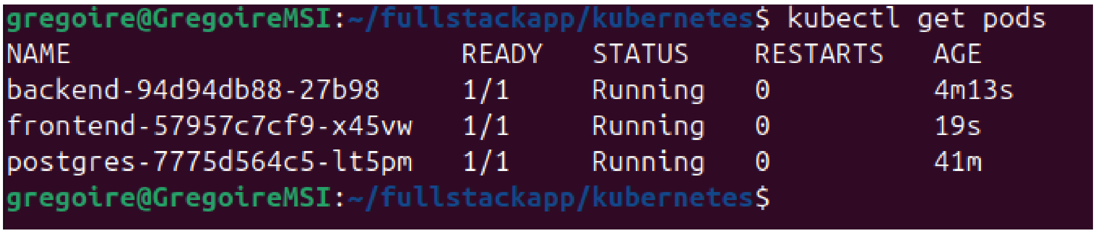
Figure 6: Pushing the backend image to the registry



```
gregoire@GregoireMSI:~/fullstackapp/frontend$ sudo docker push gregtournois/frontend
Using default tag: latest
The push refers to repository [docker.io/gregtournois/frontend]
22df3e29a025: Pushed
497f3395f0e5: Pushed
365ccd918307: Layer already exists
1bba629c69e9: Layer already exists
139c1270acf1: Layer already exists
4693057ce236: Layer already exists
latest: digest: sha256:39b4af24068d45606fb23ae52ddba54e3ed02ab4fd0e1fbbcaf8396ce02ff48d si
e: 1579
```

Figure 7: Pushing the updated frontend image

5 Successful Kubernetes Deployment



```
gregoire@GregoireMSI:~/fullstackapp/kubernetes$ kubectl get pods
NAME                        READY   STATUS    RESTARTS   AGE
backend-94d94db88-27b98     1/1     Running   0          4m13s
frontend-57957c7cf9-x45vw   1/1     Running   0          19s
postgres-7775d564c5-lt5pm   1/1     Running   0          41m
gregoire@GregoireMSI:~/fullstackapp/kubernetes$
```

Figure 8: All pods in Running state

6 Replica Strategy

The following choices were made:

- **3 frontend replicas**: stateless, cheap to duplicate, improves UI availability.
- **2 backend replicas**: avoids single point of failure but does not overload PostgreSQL

Figure 9: Three replicas for the frontend

```
GNU nano 7.2
apiVersion: apps/v1
kind: Deployment
metadata:
  name: backend
  labels:
    app: backend
spec:
  replicas: 2
  selector:
    matchLabels:
      app: backend
```

Figure 10: Two replicas for the backend

```
gregoire@GregoireMSI:~/fullstackapp/kubernetes$ kubectl get pod
NAME                         READY   STATUS    RESTARTS   AGE
backend-94d94db88-27b98      1/1     Running   0          8m30s
backend-94d94db88-2f4kx      1/1     Running   0          13s
frontend-57957c7cf9-8pqbb    1/1     Running   0          18s
frontend-57957c7cf9-gq2s8    1/1     Running   0          18s
frontend-57957c7cf9-x45vw    1/1     Running   0          4m36s
postgres-7775d564c5-lt5pm    1/1     Running   0          45m
```

Figure 11: All replicas running at the same time

7 Frontend Message Change

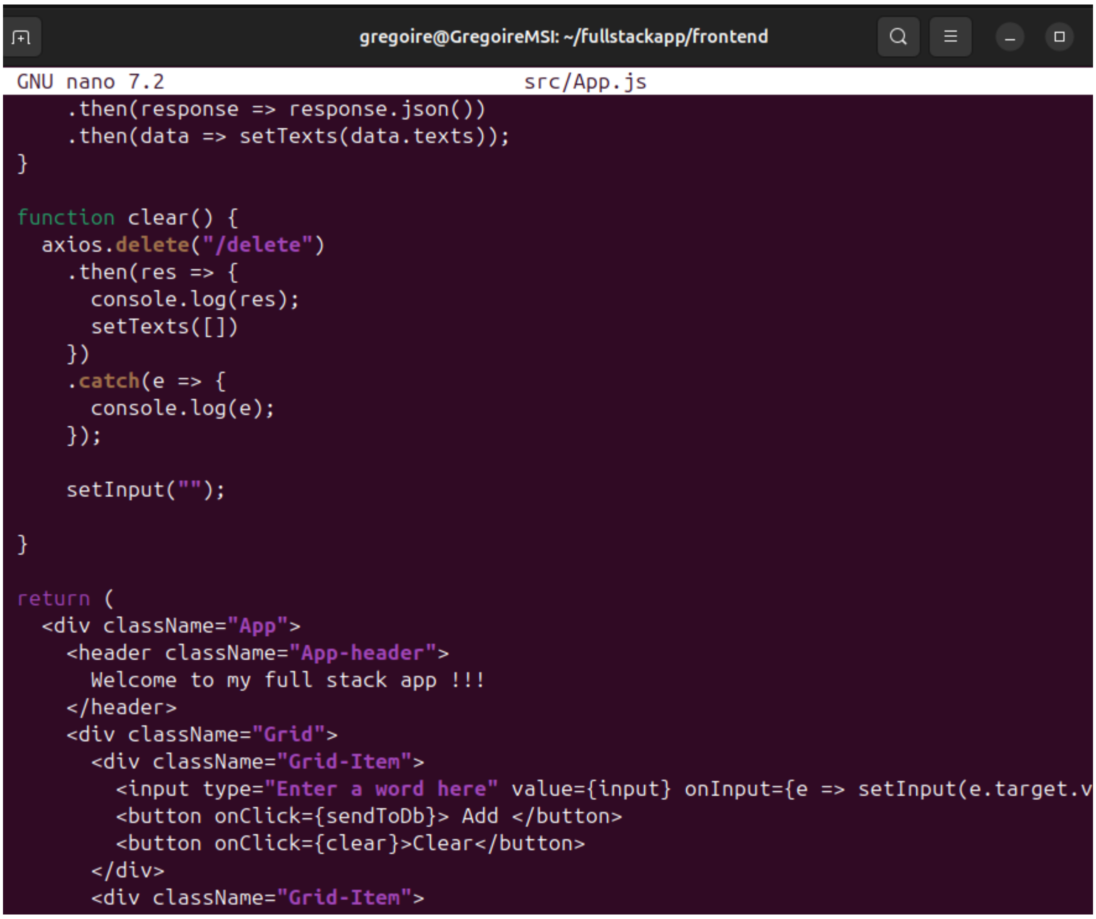


Figure 12: Edited App.js with the new message

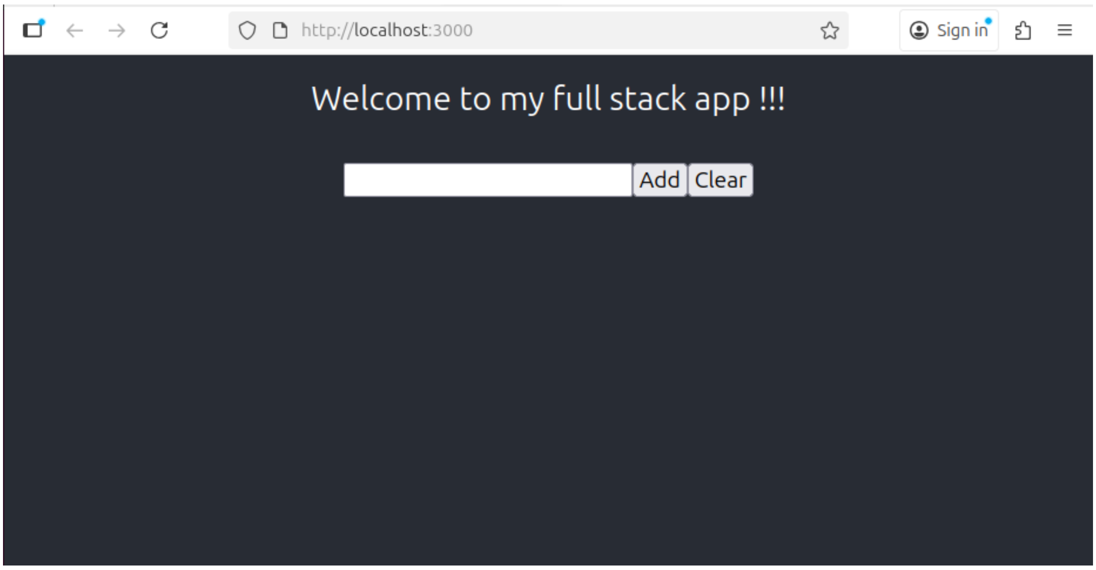


Figure 13: Frontend showing the new welcome message

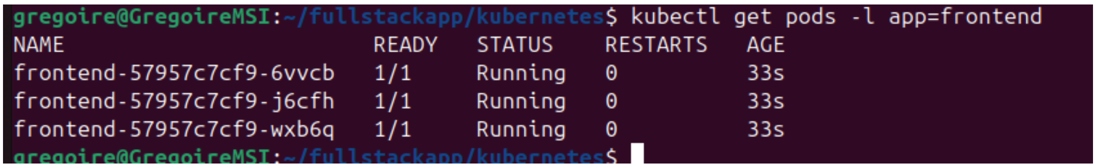


Figure 14: New frontend pods running with the updated image

8 Application Test

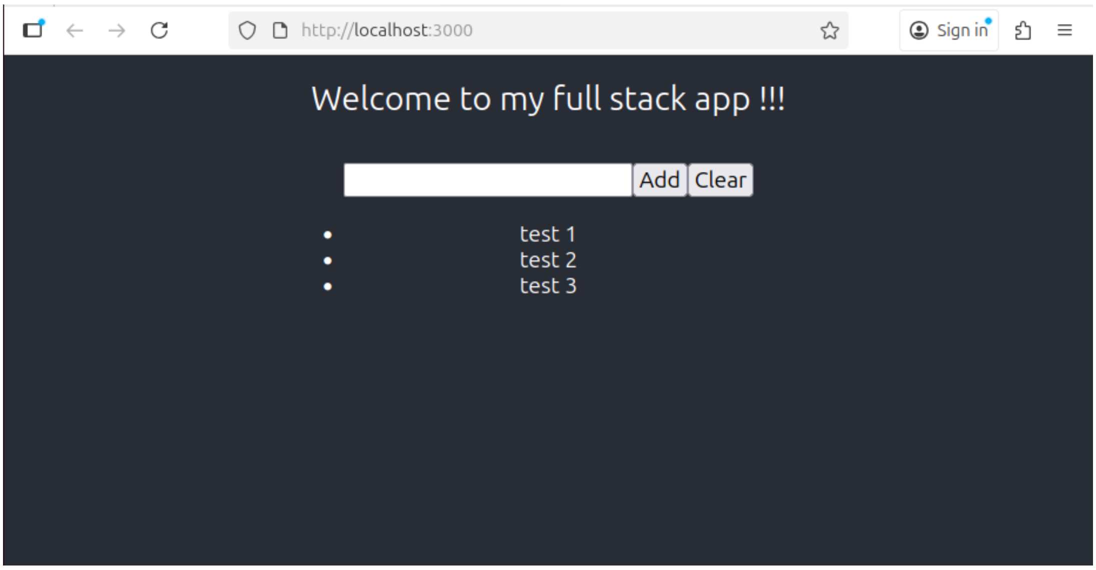


Figure 15: Access test after PostgreSQL was up